



Smart data driven traffic sign detection method based on adaptive color threshold and shape symmetry

Xianghua Xu^{a,*}, Jiancheng Jin^a, Shanqing Zhang^a, Lingjun Zhang^a, Shiliang Pu^b, Zongmao Chen^c

^a School of Computer Science, Hangzhou Dianzi University, Hangzhou 310018, China

^b Hangzhou Hikvision Digital Technology Co., Ltd., Hangzhou 310052, China

^c School of Science, Hangzhou Dianzi University, Hangzhou 310018, China

HIGHLIGHTS

- Robust and adaptive traffic sign detecting algorithms in complex traffic environment.
- Adaptive threshold traffic sign segmentation by approximate max-min normalization.
- High efficient geometrical symmetry detection with statistical hypothesis testing.
- The detection rate reached above 94% on the German traffic sign dataset GTSDb.

ARTICLE INFO

Article history:

Received 15 May 2018

Received in revised form 14 November 2018

Accepted 17 November 2018

Available online 3 December 2018

Keywords:

Traffic sign detection

Cumulative distribution function

Shape symmetry detection

GTSDb

ABSTRACT

Detection and recognition of road traffic signs constitute an important element in Advanced Driver Assistance Systems (ADAS), which can provide real-time road sign perception information to vehicles. In this paper, we proposed a new traffic sign detection method based on adaptive color threshold segmentation and the hypothesis testing of shape symmetry by leveraging traffic signs and image data. First, we calculated an adaptive segmentation threshold using the cumulative distribution function of the image histogram. Based on this, we designed an approximate maximum and minimum normalization method, which is used to suppress the interference of high brightness area and background in image thresholding processes. Secondly, we transformed the highlight shape feature of thresholding image into a connected domain feature vector. And we formulated a shape symmetry detection algorithm based on statistical hypothesis testing to efficiently extract the ROI of traffic signs based on traffic data analysis. We performed some comprehensive experiments on the GTSDb (German Traffic Sign Detection Benchmark) dataset. The accuracy of traffic sign detection exceeded 94%. This method has higher detection accuracy and time efficiency than other methods, and better robustness under complex traffic environment.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Detection and recognition of road traffic sign constitute an important technological element in Advanced Driver Assistance Systems [1] (ADAS), which can provide real-time road sign information to vehicles and enhance driving safety [2]. Traffic signs are usually composed of specific shapes (round, square, and triangular) and colors (red, blue, and yellow), which have significant visual effect in the road environment. Therefore, traffic sign detection can be classified into color-based, shape-based and color-shape-based methods [3,4].

In color-based traffic sign detection, the RGB images are usually transformed into other color space such as HSI [5], CIELab [6], and

HSL [7]. Then the traffic signs are extracted by color thresholding segmentation based on smart data processing. Color-based traffic sign detection methods are easily affected by complex illumination conditions of traffic scenes. In shape-based traffic sign detection, geometric contour shapes of traffic signs are detected by means of template matching [8], geometric invariant moment [9], and geometric symmetry [6,10]. Symmetry detection has better adaptability compared with template matching, geometric invariant moment in a complex illumination environment, but it requires higher computational complexity.

The existing color and shape based traffic sign detection methods have weaker adaptability under complex brightness conditions. In this paper, we propose an adaptive color threshold segmentation and high efficient shape symmetry algorithm to achieve robust traffic sign detection in a complex illumination environment. (1) In color thresholding segmentation: we calculate an

* Corresponding author.

E-mail address: xhxxu@hdu.edu.cn (X. Xu).

adaptive threshold using the cumulative distribution function of image histogram; and we use the approximate maximum–minimum normalization method to process images according to the threshold. With this method, the interference of overexposure foreground objects and background is suppressed, so we can achieve better traffic sign segment effect in complex lighting environments. (2) In geometrical symmetry detection: by converting connected regions of a binary image into shape geometrical feature vectors, we use the statistical hypothesis testing method to judge the symmetry of connected components in a binary image, and extract symmetry components as the candidate ROIs of traffic signs. With this method, we realize higher efficient geometrical symmetry computation of the image than with the previous algorithm based on frequency domain.

The traffic sign detection process is shown in Fig. 1. The main contributions of this paper are included in the second and fifth steps (detailed algorithms are presented in Section 3).

Step 1: Traffic images are transformed into Red–Blue gray images by Red–Blue normalization processing [11];

Step 2: The adaptive segmentation threshold is calculated based on the cumulative distribution function of the histogram of Red–Blue images; and the images' foreground enhancement is processed using the approximate maximum–minimum normalization method to suppress the images' high brightness background;

Step 3: Candidate connected domains of traffic signs are extracted by morphological filtering and MSER [11];

Step 4: Enhanced processing of the connected domains is performed using convex operation;

Step 5: Symmetry statistical analysis and hypothesis testing are performed on the feature vectors of connected domains to detect shape symmetry and extract candidate ROIs of traffic signs.

Step 6: Traffic signs' ROI is determined using geometric constraints.

The paper is organized as follows: Section 2 describes related works about traffic sign detection, Section 3 details the main contributions of this paper, Section 4 shows some experimental results of our method, and Section 5 presents the conclusions.

2. Related works

Traffic signs have significant color and shape features. As shown in Fig. 2, the GTSRB traffic sign dataset [12] includes prohibition signs (circular with a red outer ring), indication signs (circular with blue background) and warning signs (red upper triangle). Traffic sign detection methods are mainly classified into: color information based, shape information based, and fusion of color and shape information based detection algorithms.

Color information based traffic sign detection [13–16]: Color information is a prominent feature of traffic signs, with size and perspective invariability. Hence, the ROI of traffic signs can be quickly located based on color information. They generally use color threshold segmentation in a specific color space to extract traffic sign ROI. Escalera [17] sets the thresholds of RGB color channels to perform segmentation of RGB images. Since the RGB color space is sensitive to image brightness, the segmentation effect is easily affected by brightness conditions of a complex traffic environment. Therefore, more methods are generally used to transform the images from RGB color space to other color space. Gao [18] segmented images on Hue and Chroma values of the CIECAM97 color model with quadtree histograms of the image. Creusen [14] extracted the HOG feature on YcbCr color space to detect traffic signs. Greenhalgh [11] proposed the Red–Blue preprocessing method to perform color normalization on the red and blue channels of images to reduce the influence of complex brightness conditions.

Shape information based traffic sign detection: Shape information is another prominent feature of traffic signs. Shape-based traffic sign detection methods generally use contour feature to extract traffic sign's ROI. F. Moutard [19] detected round and rectangular traffic signs by Hough transform and Canny edge detection. Escalera [20] used template matching of the signs' contour to detect traffic signs. Traffic signs also have obvious geometrical symmetry. Xu [10] used shape symmetry detection methods to extract circular and triangular signs. Keller [21] used radial symmetric voting to extracting shape information, and then used Harr wavelet feature based cascade classifier to detect traffic signs. Wang [22] detected radial symmetry based on local affine invariant edge response. Although shape information based traffic sign detection is more adaptable to complex brightness conditions, it requires higher computing complexity and the performance is influenced greatly by deformation, rotation, and partial occlusion of traffic signs.

Color and shape information based traffic sign detection: Color and shape features are combined into traffic sign detection algorithms to enhance detection performance. Zhu [23] integrated color and shape features of traffic signs into an energy expression form to detect traffic signs. Deguchi [24] used a nested cascade classifier based on local rank pattern feature to classify RGB pixel values to obtain edge images for seven traffic signs, and then used RANSAC circular fitting to detect circular traffic signs. The detection method proposed in [15,25–28] had better robustness in a complex road environment. However, most of the existing methods are only limited to specific types of traffic signs. For example, in [26,27] only red circular signs were detected. In addition, these methods generally were evaluated with the authors' own traffic sign dataset rather than the public traffic sign dataset.

In summary, the existing color and shape based traffic sign detection methods have weaker adaptability under complex brightness conditions. Hence, we propose a novel adaptive color threshold segmentation and statistical hypothesis test symmetry detection algorithm to enhance traffic sign detection performance under complex illumination environments.

3. Adaptive color threshold and shape symmetry testing in traffic sign detection

In this paper, we extract traffic signs' regions (ROIs) through some prior fusion information of color and shape of traffic signs. This method has better adaptability to complex scenes such as illumination, rotation and scale change in road traffic environments.

3.1. Adaptive threshold segmentation based on an Approx. Max–Min normalization

In order to suppress the interference of highlight areas and enhance color contrast of traffic signs in traffic images, we propose a method of determining the adaptive threshold of color segmentation for traffic signs, and an algorithm of image enhancement processing by using approximate maximum–minimum normalization. With this method, traffic signs can be significantly enhanced relative to the background in traffic images.

Algorithm idea: First, a traffic image is transformed into a Red–Blue gray scale image by Red–Blue normalization processing. Then, its color threshold value is determined by the cumulative distribution function of its image gray histogram. And an approximate minimum–maximum normalization processing is performed on this image according to its predetermined threshold. Finally, the color segmentation is completed with morphological filtering and MSER method. In this way, most of the invalid backgrounds can be filtered to highlight the foreground color blocks, while the overexposed areas are smoothed out to reduce the influence of high contrast and brightness.

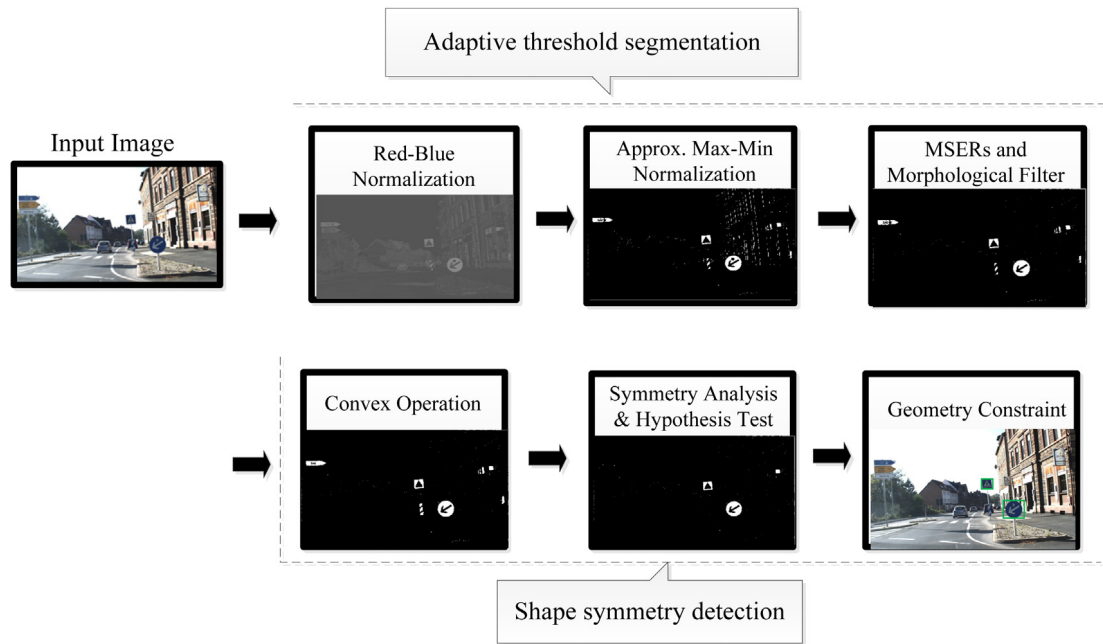


Fig. 1. Flow chart of the traffic sign detection process. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



Fig. 2. Traffic sign samples in GTSD dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The image after Red-Blue processing is still affected by high contrast and brightness in a complex traffic environment. As a result, the foreground sign may not be highlighted in a highly contrasting and highly bright image. In order to solve these problems, we propose an adaptive threshold segmentation method called approximate maximum–minimum normalization:

(1) Cumulative distribution function of a Red-Blue image: Noting the Red-Blue image of a detection sample as $RB(x)$, which has a gray-scale distribution probability density function $p(k)$ (as shown in Fig. 3). The cumulative distribution function (CDF) is $F(m)$:

$$F(m) = \sum_{k=\min}^m p(k) \quad (1)$$

wherein m is a specified grayscale value to be determined.

(2) Adaptive threshold selection based on the cumulative distribution function of the Red-Blue image: We have observed that the gray values of the foreground region are often distributed within a certain range by statistical analysis experiments. Therefore, an adaptive threshold can be determined according to the following solution:

Finding $m = m_1$ to satisfy

$$\begin{cases} F(m_1 - 1) < \beta & \text{and} \\ F(m_1) \geq \beta \end{cases} \quad (2)$$

Wherein $\beta \in [0, 1]$ (Setting $\beta = 0.9$ in this paper). The pixel value m_1 , which corresponds to β , is the adaptive threshold for the gray image $RB(x)$.

(3) Approximate maximum–minimum normalization: Normalize the Red-Blue image $RB(x)$ according to the adaptive threshold

m_1 :

$$RB'(x) = \begin{cases} 0, & RB(x) < m_1 \\ 255, & RB(x) = m_1 \\ \frac{RB(x) - m_1}{\max(RB) - m_1} * 255, & RB(x) > m_1 \end{cases} \quad (3)$$

Through the above process, most of the background is eliminated, so as to highlight the foreground color blocks. The overexposed areas are also suppressed to reduce the influences of low contrast, high brightness and other environmental factors. The processing results of adaptive threshold segmentation under low and high brightness environments are shown in Fig. 4. The left column contains the original images; Middle column contains Red/Blue normalized images; Right column contains the segmentation results of approximate maximum–minimum normalization with an adaptive threshold. Experiments results validated our method's adaptability to low and high brightness environments.

Traffic signs' area can be effectively enhanced by approximate maximum–minimum normalization with an adaptive threshold. And most of the background is suppressed. However, there are still some noises and isolated points in the resulting images, so more accurate ROI candidates of traffic signs are obtained by further morphological filtering and MSER processing. As shown in Fig. 5, our adaptive threshold segmentation method has high robustness in complex traffic scenes such as high contrast, high brightness, multiscale and various illumination conditions.

3.2. Shape feature vector based symmetry detection

For geometric symmetry detection, we transform the connected regions in a binary image into feature vectors, and use statistical

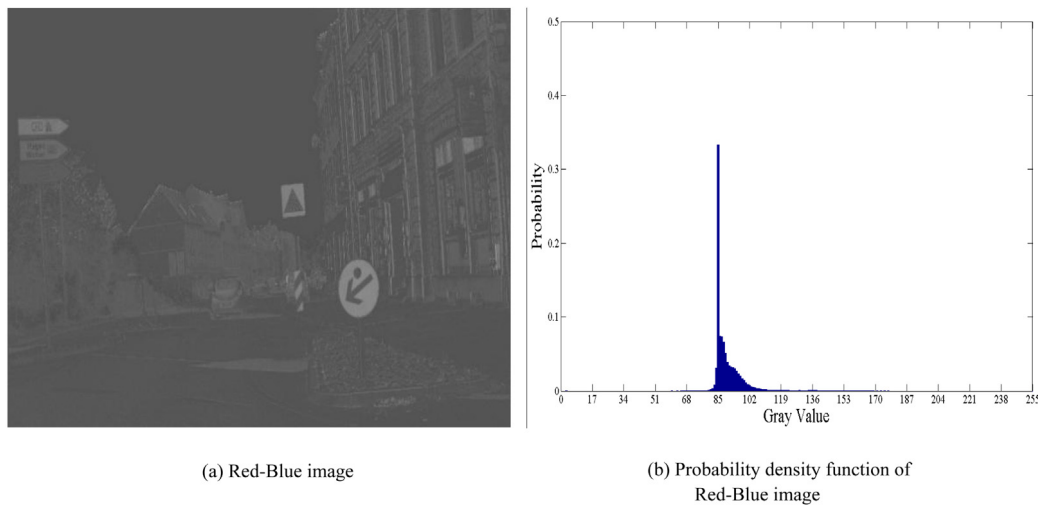


Fig. 3. Probability histogram of a gray Red-Blue image. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

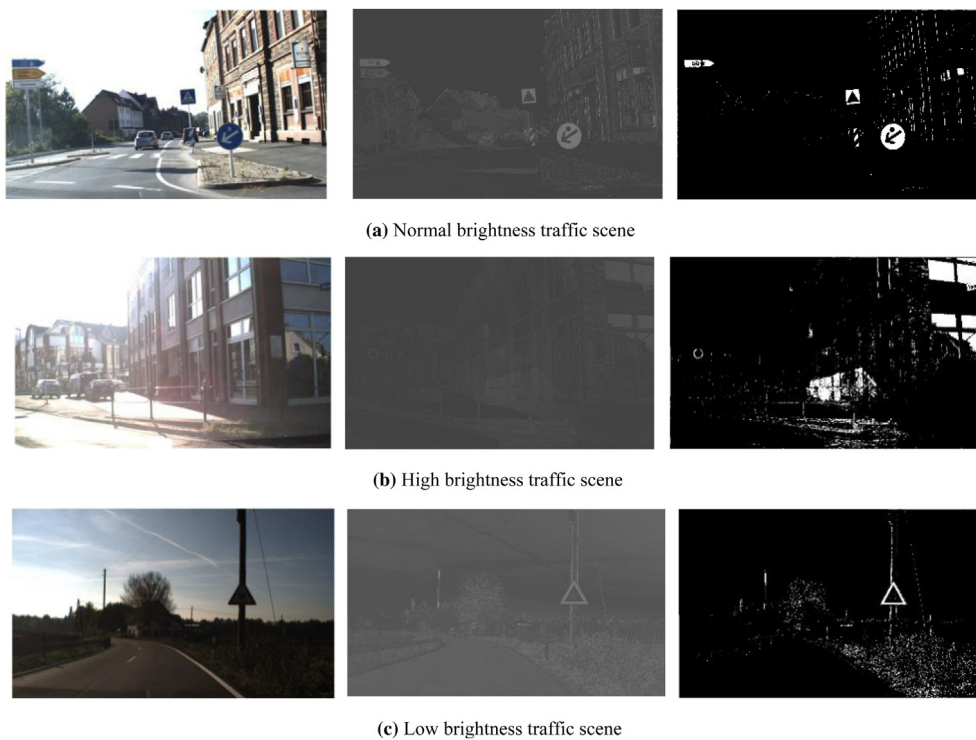


Fig. 4. Image processing results: Left column contains the original images; Middle column contains Red/Blue normalized images; Right column contains the resulting images that are processed by approximate maximum-minimum normalization with an adaptive threshold. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

hypothesis testing to judge geometric symmetry of the connected components. Then, ROI candidates with symmetry features are preserved. This algorithm realizes real-time shape symmetry detection.

After the color features are extracted, the overall area of the traffic signs will be retained at a large extent. However, in complex traffic scenes, defects of edge contour after MSER processing may occur due to poor lighting conditions, low contrast, fading and occlusion. This will greatly affect the subsequent symmetry detection. Hence, convex operation [10] is used to determine convex connected components to avoid such errors as far as possible. The result of the convex operation is shown in Fig. 6.

After convex operation, the symmetry of the connected components can be verified by the symmetry detection method in [5,29], but this algorithm has low efficiency because it needs to convert the image from spatial domain to frequency domain. Through experiments, it has been found that the statistical data of symmetric graphs also keep symmetry. As shown in Fig. 7, the histogram of a triangle will still appear as a triangle, while that of a circle will appear as a semicircle. Asymmetrically connected components still exhibit asymmetrical statistics data, as shown in Fig. 8.

Based on the above observation, we propose a feature vector describing symmetry of a region as shape features of traffic signs, thereby transforming detection of the shape symmetry into statistical hypothesis testing of the feature vector.



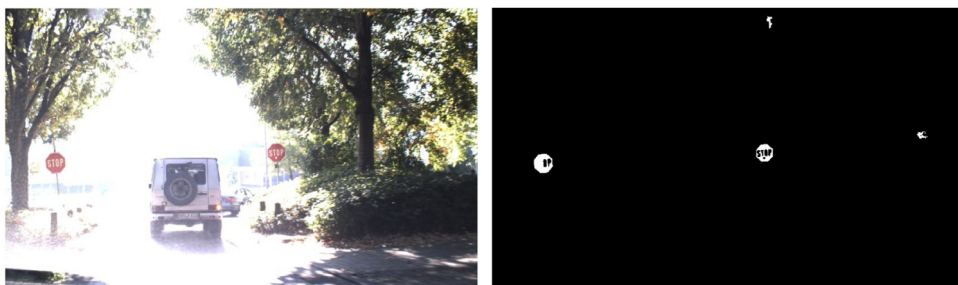
(a) Multiscale scene of traffic signs



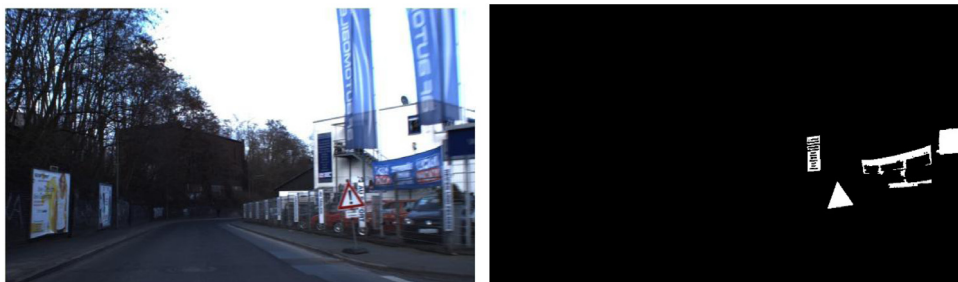
(b) High contrast scene of traffic signs



(c) Multiple traffic signs close to each other



(d) Traffic signs of different brightness in a highly contrasting background

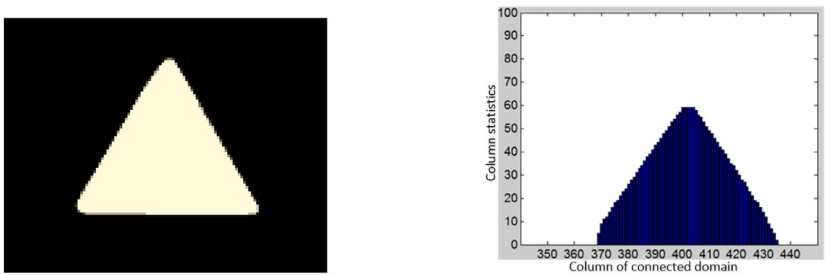


(e) Rotated traffic signs

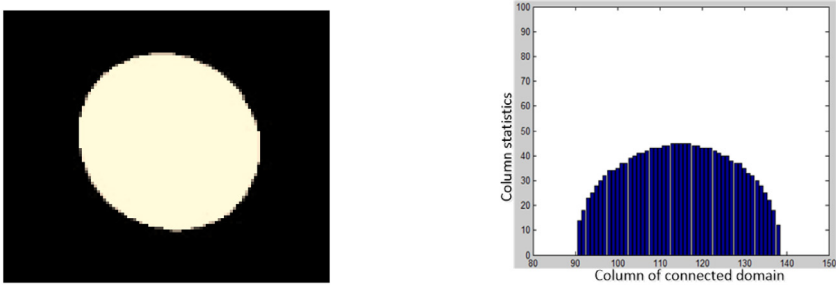
Fig. 5. Results of MSER filtering. Left column contains the original images; Right column contains the corresponding resulting images after MSER filtering.



Fig. 6. Convex operation result. Left column is the original image. Middle column is the binary image after MSER. Right column is the binary image after convex operation.



(a) Histogram of the connected domain of a triangle



(b) Histogram of the connected domain of a circle

Fig. 7. Statistical histograms of symmetrical shapes in a binary image.

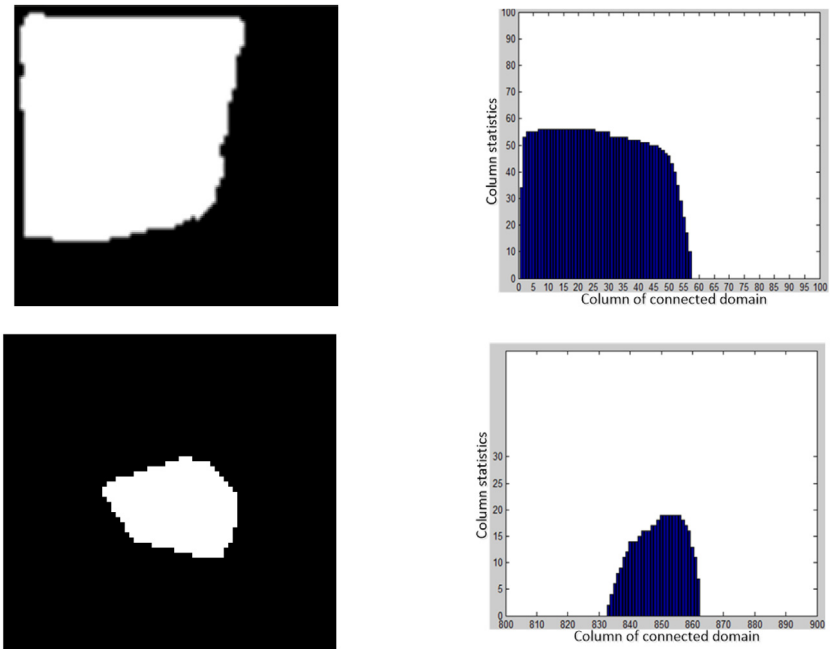


Fig. 8. Statistical histograms of asymmetrical shapes in a binary image.

Shape symmetry detection algorithm: First, we perform column projection on connected components and count the number of effective pixels in each column. Then, we use the Minkowsky subtraction to construct a feature vector of the connected components. It is evident that the stronger the symmetry of a connected component is, the mean value and variance of its feature vectors are closer to 0. Hence, the hypothesis test will be established for means of feature vectors' samples. Finally, the symmetry of the connected component is determined by the hypothesis test on its feature vectors. The shape symmetry detection algorithm is as follow:

(1) Column Projection of Convex Connected Component.

The number of the effective pixels in each column of a convex connected region is denoted by C_i ($i = 1, 2, 3, \dots, N$), wherein N is the total number of columns in the convex connected component;

(2) Feature vector of a convex connected component constructed using the Minkowski subtraction [30].

If a region is strictly symmetric, it should satisfy $C_i = C_{N+1-i}$. But in a real scene, due to lens distortion, inaccurate image segmentation, there is some pixel position offset. Therefore, it is more appropriate that the strict corresponding column and its front and back columns are used as reference in the processing. We use a grayscale calculation method based on the Minkowski subtraction to solve the offset error problem and define:

$$\begin{cases} d_1 = \min\{|C_1 - C_N|, |C_1 - C_{N-1}|\} \\ d_2 = \min\{|C_2 - C_{N-2}|, |C_2 - C_{N-1}|\} \\ \dots \dots \\ d_{\frac{N}{2}} = \min\left\{|C_{\frac{N}{2}} - C_{N-\frac{N}{2}}|, |C_{\frac{N}{2}} - C_{N+1-\frac{N}{2}}|, |C_{\frac{N}{2}} - C_{N+2-\frac{N}{2}}|\right\} \end{cases} \quad (4)$$

Wherein d_i is the minimum difference between the i th column and the N th column, the $(N+1)$ th column, and the $(N+2)$ th column (special when $i = 1$), and these difference values are concatenated as a feature vector of a connected component: $d = \{d_1, d_2, d_3, \dots, d_{\frac{N}{2}}\}$.

(3) Hypothesis testing on feature vector space.

Having calculated the mean and variance of the feature vector obtained from each connected component, we find that the stronger the symmetry is, the closer the mean and variance are to 0. However, in an actual scene, due to the complexity of the image itself and errors of detection methods, the feature vector is not completely symmetrical. Considering the independent and identical distribution characteristics of feature vector data, some asymmetrically connected components can be filtered by hypothesis testing, and the symmetrically connected components can be preserved.

Suppose the feature vector of a connected component is independent and identically distributed, and D is the overall feature vectors constructed from traffic signs, and $d_1, d_2, d_3, \dots, d_n$ ($n = N/2$) are samples in D , the mean and variance of the samples are calculated as shown in Eq. (5).

$$\begin{cases} \bar{d} = \frac{d_1 + d_2 + \dots + d_n}{n} \\ S^2 = \frac{(d_1 - \bar{d})^2 + (d_2 - \bar{d})^2 + \dots + (d_n - \bar{d})^2}{n-1} \end{cases} \quad (5)$$

According to statistics principles [31], when the volume of the sample data is larger (generally greater than 45), the statistics $\frac{\bar{d} - \mu_0}{\sigma/\sqrt{n}}$ generally approaches normal distribution, and follows t distribution when the volume of the sample data is smaller. Therefore, the following hypothesis test can be established:

(a) When the volume of the sample data is smaller, the statistic $\frac{\bar{d} - \mu_0}{\sigma/\sqrt{n}}$ approximately obeys t -distribution. According to the characteristics of symmetry and actual error and experimental error, two

Table 1

Geometric constraints used to screen candidate traffic signs' ROI.

Geometric constraints	Minimum(pixel)	Maximum(pixel)
Width	12	130
Height	12	130
Height/Width	0.8	1.2
Area pixels	144	16900

mutually opposing assumptions are proposed.

$$\begin{cases} H_0: \mu \leq \mu_0 \\ H_1: \mu > \mu_0 \end{cases} \quad (6)$$

Wherein μ_0 is the overall expectation, and σ, n represent the standard deviation and the number of samples, respectively. In addition, since σ^2 is unknown, we cannot use $\frac{\bar{d} - \mu_0}{\sigma/\sqrt{n}}$ to determine the rejection domain (rejection domain form is $t = \frac{\bar{d} - \mu_0}{S/\sqrt{n}} \geq k$, wherein k is a positive constant), and S^2 is an unbiased estimate of σ^2 , so replace σ with S , we use $t = \frac{\bar{d} - \mu_0}{S/\sqrt{n}}$ as the statistic for the test, there is $\frac{\bar{d} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$, so

$$P\{\text{When } H_0 \text{ is true, reject } H_0\} = P_{\mu_0} \left\{ \frac{\bar{d} - \mu_0}{\frac{S}{\sqrt{n}}} \geq k \right\} = \alpha \quad (7)$$

In the one-side test, $k = t_{\alpha}(n-1)$, that is, there is a rejection domain $t = \frac{\bar{d} - \mu_0}{S/\sqrt{n}} \geq t_{\alpha}(n-1)$, wherein the significant level α is 0.05, then you can obtain the value of $t_{\alpha}(n-1)$ by looking it up in the table. This means that when 95% of the samples fall within the confidence interval, the overall area of the corresponding sign can be determined as a symmetrically connected domain, otherwise as an asymmetrically connected domain.

(b) When the volume of the sample data is larger, the statistic $\frac{\bar{d} - \mu_0}{\sigma/\sqrt{n}}$ approximately obeys normal distribution, there is $\frac{\bar{d} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$, wherein μ_0, σ, n respectively represent the overall expectation, the population standard deviation and the number of samples. Replacing σ with S , the statistic $z = \frac{\bar{d} - \mu_0}{S/\sqrt{n}}$ is tested. Similarly, setting the significance level $\alpha = 0.05$, we can get the rejection domain form under normal distribution:

$$P\{\text{When } H_0 \text{ is true, reject } H_0\} = P_{\mu \in H_0} \left\{ \frac{\bar{d} - \mu_0}{\frac{S}{\sqrt{n}}} \geq k \right\} = \alpha \quad (8)$$

That is, the rejection domain of (8) is $z = \frac{\bar{d} - \mu_0}{S/\sqrt{n}} \geq z_{\alpha}$, wherein you can obtain z_{α} by looking it up in the table. The significance level α is 0.05, meaning that when 95% of the samples fall within the confidence interval, the overall area of a sign can be determined as a symmetrically connected domain, otherwise it is an asymmetrically connected domain.

The above method converts symmetry detection of a connected domain to hypothesis testing of the corresponding feature vector. The experimental results are shown in Fig. 9. Our method has a significant detection effect on traffic signs, effectively reduces noise interference, and can be used to distinguish traffic signs from environmental background, has good robustness to local contrast changes and overall brightness contrast etc. Compared with other detection methods, it has better real-time performance.

Symmetry detection largely eliminates invalid background areas, preserves traffic signs and a small number of regional blocks similar in color and shape to traffic signs. We finally perform geometric constraints of height, width, ratio of traffic signs on the candidate regional blocks to further screen the traffic signs' ROI region. The geometric constraints used for screening in the experiment are shown in Table 1.

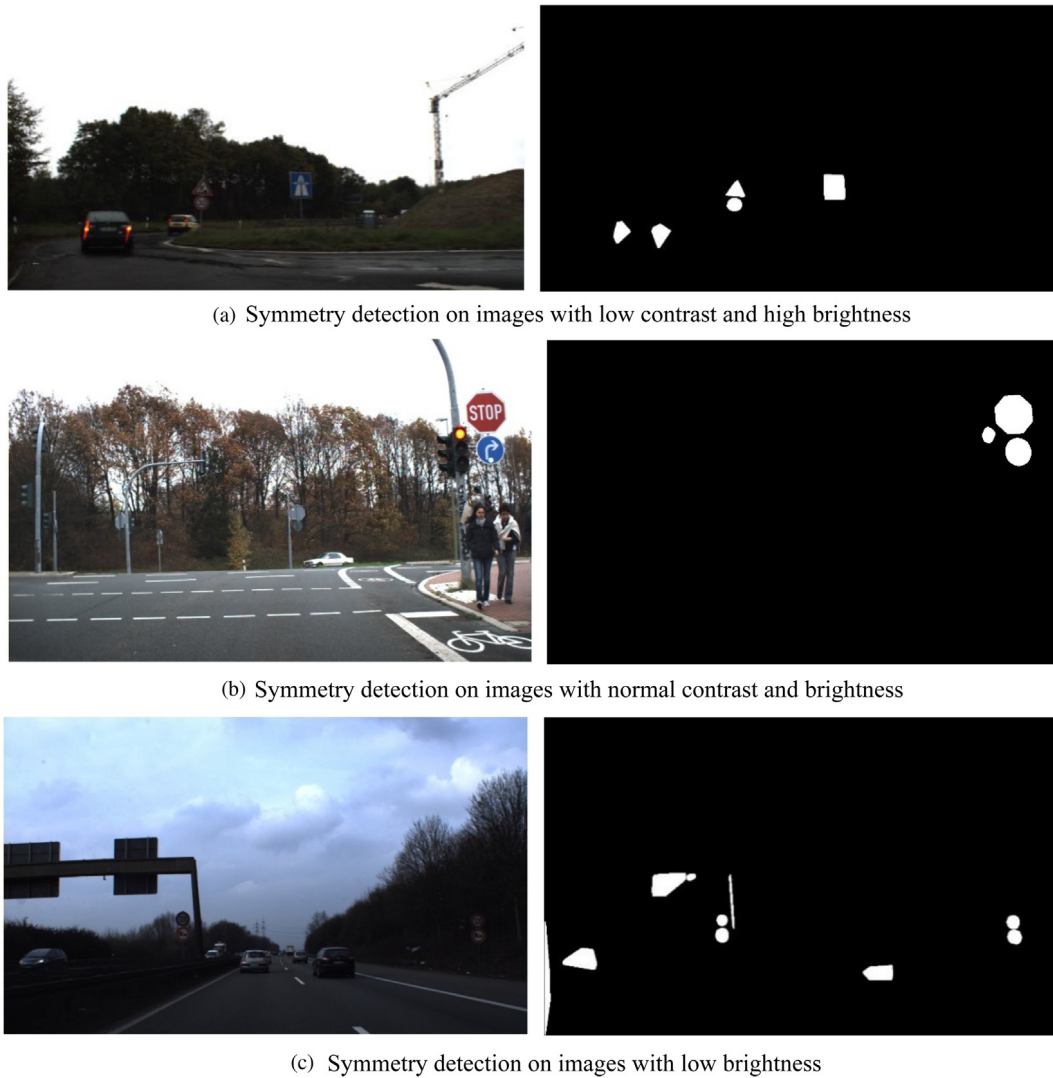


Fig. 9. Symmetry detection in different brightness and contrast traffic scene.

4. Experiment results

4.1. Traffic sign dataset

German Traffic Sign Detection Benchmark (GTSDDB) is a public traffic sign dataset. The dataset has 900 images with size of 1360×800 pixels (where 600 images are images for training and others are images for testing). The dataset contains various road scenes with dramatic changes in brightness and contrast conditions. The sizes of traffic signs' candidate ROIs on the GTSDDB range from 16×16 to 128×128 . In the experiments, the traffic signs are manually marked on the images to facilitate statistical evaluation of the algorithm's detection performance.

4.2. Experiment results of traffic sign detection

In experiments, our traffic sign detection algorithm is compared with Jack Greenhalgh's method [11] and MA Garcia-Garrido's method [32]. Jack Greenhalgh used MSER to detect candidate ROI of traffic signs. MA Garcia-Garrido used threshold image segment and Hough transform to detect candidate ROI of traffic signs. Table 2 shows the comparative experimental results on the same dataset.

Matching the minimum-bounding rectangle ROI from the experiment with the manually calibrated ROI rectangle, we used Jaccard's similarity coefficient [12] to determine whether the detected

Table 2

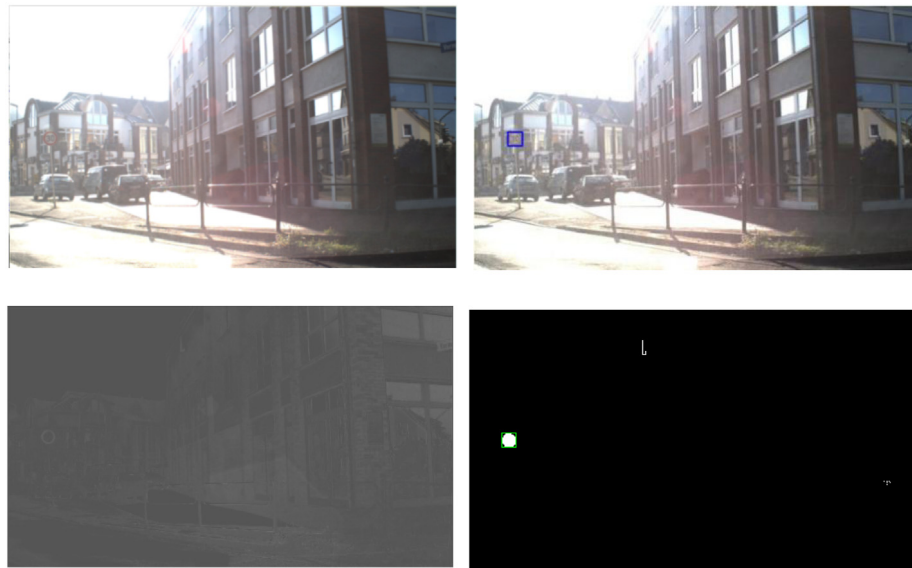
Traffic signs detection results on the GTSDDB dataset.

Detection method	Our method	Jack Greenhalgh [11]	MA Garcia-Garrido [32]
Total number of traffic signs	719	719	719
Number of candidate areas	729	915	868
Number of correct detections	685	632	670
Number of false alarms	10	196	149
Accuracy	93.96%	69.07%	77.19%
Recall rate	95.27%	87.90%	93.18%
F-measure	94.61%	77.36%	84.43%

ROI meets the detection requirements of the actual ROI region. The evaluation coefficient is described in Eq. (9):

$$J(S, G) = \frac{|S \cap G|}{|S \cup G|} \in [0, 1] \quad (9)$$

Wherein S represents the ROI block that is actually detected, and G represents the ground-truth of ROI block. When Jaccard similarity coefficient J exceeds a threshold, the ROI is considered



(a) Traffic sign detection results of high brightness conditions



(b) Traffic sign detection results of multiscale traffic signs

Fig. 10. Illustrate the robustness of our traffic sign detection methods. The image at the upper left corner is the original image; the image at the upper right corner is our detection results; the image on the lower left corner is detection results with Jack Greenhalgh's method; the image on the lower right corner is detection results with MA Garcia-Garrido's method.

Table 3
Average ROI results on the GTSDDB dataset.

Detection method	Our method	Jack Greenhalgh	MA Garcia-Garrido
Average J	81.3%	76.5%	75.9%

Table 4
Average time for traffic sign detection on the GTSDDB dataset.

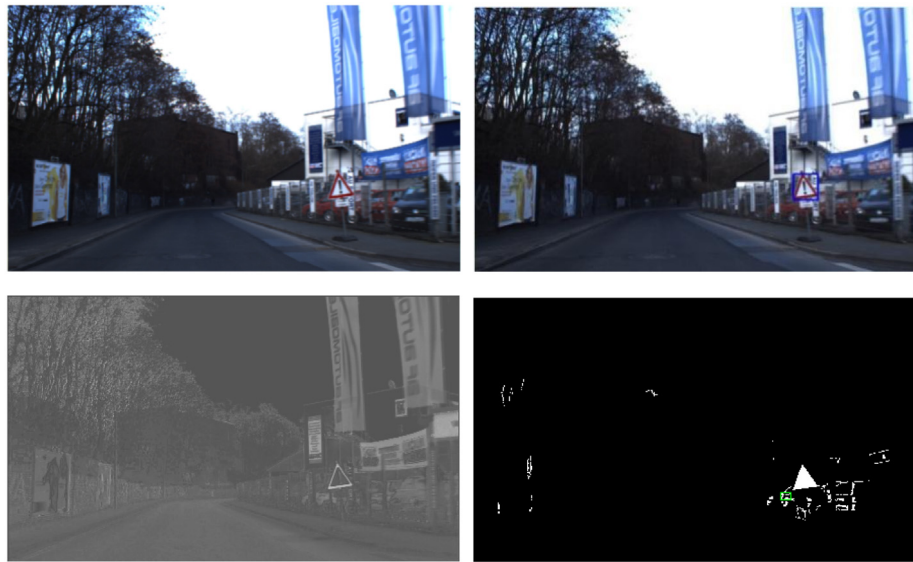
Detection method	Proposed method	Escalera [8]	MA Garcia-Garrido [32]
Average time (s)	0.930	2.035	1.680
shape detection (s)	0.55	1.25	0.95

to meet the detection requirements; otherwise, the ROI is not counted in the detection result. The average Jaccard similarity coefficient J obtained by three methods is shown in Table 3:

The experimental results in Tables 2 and 3 show that our traffic sign detection algorithm has significantly better performance than algorithms in [11] and [32].

In addition, the run time of the three methods are shown in Table 4. In [8], Escalera adopted the template matching method

following color segmentation to filter out the ROI blocks that did not meet symmetry conditions. From Table 4, it can be seen that shape verification in the traffic sign detection process consumed the major amount of the time used for detection. Therefore, how to decrease the number of objects for shape detection is a key to improving the overall detection speed. Our method improves the detection speed in three ways, including the use of adaptive



(c) Traffic sign detection results of high contrast conditions



(d) Traffic sign detection results of low contrast conditions

Fig. 10. (continued).

threshold segmentation to filter out most invalid backgrounds, the MSER process before shape detection, and symmetry detection on spatial domain instead of frequency domain.

Fig. 10 shows traffic sign detection results on the GTSDDB dataset. Compared with Jack Greenhalgh's and MA Garcia-Garrido's method, our algorithm has higher detection performance under different brightness, contrast, scale changes and rotation conditions. In Fig. 10, the image at the upper left corner is the original image; the image at the upper right corner is our detection results; the image on the lower left corner is Jack Greenhalgh's detection results; the image on the lower right corner is MA Garcia-Garrido's detection results. As shown in Fig. 10(a)–(d), our method has correctly detected the traffic sign in complex traffic scene, such as high brightness, high contrast, low contrast and multiscale conditions. Our method has higher robustness compared with Jack Greenhalgh's and MA Garcia-Garrido's method in complex brightness traffic scene.

5. Conclusions

This paper presents a new traffic sign detection method based on adaptive color threshold segmentation and shape symmetry hypothesis testing by leveraging traffic signs and image data. First, in the color segmentation stage, the cumulative distribution function of the histogram is used to dynamically determine the segmentation threshold. And the approximate maximum–minimum normalization method is used to suppress the interference by high brightness area and background. So, the threshold segment algorithm has adaptability for images of different traffic scenes. Second, the highlight shape feature of the threshold image is transformed into a connected domain feature vector. And the statistical hypothesis testing based shape symmetry detection algorithm is formed to extract the ROI of traffic signs efficiently. Through testing on the German traffic sign dataset GTSDDB, the detection rate exceeded 94%, which is higher than those of the

methods in [11,32]. The detection method has good robustness to variations in brightness, contrast, scale and rotation in complex traffic environments.

Acknowledgments

This work is supported by the Key Science-Technology Program of Zhejiang Province, China (2017C01022) and the National Natural Science Foundation of China (61370087).

References

- [1] Z. Malik, I. Siddiqi, Detection and recognition of traffic signs from road scene images, in: International Conference on Frontiers of Information Technology, 2015.
- [2] L. Zhou, Z. Deng, LIDAR and vision-based real-time traffic sign detection and recognition algorithm for intelligent vehicle, in: IEEE International Conference on Intelligent Transportation Systems, 2014.
- [3] C. Liu, F. Chang, Z. Chen, Rapid multiclass traffic sign detection in high-resolution images, *IEEE Trans. Intell. Transp. Syst.* 15 (6) (2014) 2394–2403.
- [4] A. Mogelmose, D. Liu, M.M. Trivedi, Traffic sign detection for US roads: Remaining challenges and a case for tracking, in: IEEE International Conference on Intelligent Transportation Systems, 2014.
- [5] J.M. Lillo-Castellano, I. Mora-Jiménez, C. Figuera-Pozuelo, et al., Traffic sign segmentation and classification using statistical learning methods, *Neurocomputing* 153 (2015) 286–299.
- [6] S. Sooksattra, T. Kondo, Red traffic light detection using fast radial symmetry transform, in: International Conference on Electrical Engineering/electronics, Computer, Telecommunications and Information Technology, 2014.
- [7] M. Kehl, M. Enzweiler, B. Froehlich, et al., Vision-based road sign detection, in: IEEE International Conference on Intelligent Transportation Systems, 2015.
- [8] S. Escalera, X. Baró, O. Pujol, et al., *Background on Traffic Sign Detection and Recognition*, Springer, London, 2011, pp. 4–14.
- [9] X. Ma, C. Mu, Y. Wang, et al., Traffic sign detection and recognition using color standardization and Zernike moments, in: Chinese Control and Decision Conference, 2016.
- [10] Z. Xu, J. Ren, C. Bao, A Traffic signs' detection method of contour approximation based on concave removal, in: Chinese Control and Decision Conference, 2016.
- [11] J. Greenhalgh, M. Mirmehdi, Real-Time detection and recognition of road traffic signs, *IEEE Trans. Intell. Transp. Syst.* 13 (4) (2012) 1498–1506.
- [12] S. Houben, J. Stallkamp, J. Salmen, et al., Detection of traffic signs in real-world images, in: The German traffic sign detection benchmark. International Joint Conference on Neural Networks, 2014.
- [13] M. Benallal, J. Meunier, Real-time color segmentation of road signs, in: IEEE Canadian Conference on Electrical and Computer Engineering, 2003.
- [14] I.M. Creusen, R.G.J. Wijnhoven, E. Herbschleb, et al., Color exploitation in hog-based traffic sign detection, in: IEEE International Conference on Image Processing, 2010.
- [15] H. Li, F. Sun, L. Liu, et al., A novel traffic sign detection method via color segmentation and robust shape matching, *Neurocomputing* 169 (2015) 77–88.
- [16] L.W. Tsai, J.W. Hsieh, C.H. Chuang, et al., Road sign detection using eigen colour, *Int. Comput. Vis.* 2 (3) (2008) 164–177.
- [17] L.E.A. De, L.E. Moreno, M.A. Salichs, et al., Road traffic sign detection and classification, *IEEE Trans. Ind. Electron.* 44 (6) (1997) 848–859.
- [18] X. Gao, K. Hong, P. Passmore, et al., Colour vision model-based approach for segmentation of traffic signs, *Eurasip J. Image & Video Process.* (1) (2008) 1–7.
- [19] F. Moutarde, A. Bargeton, A. Herbin, et al., Robust on-vehicle real-time visual detection of American and European speed limit signs, with a modular Traffic Signs Recognition system, in: IEEE Intelligent Vehicles Symposium, 2007.
- [20] S. Escalera, X. Baró, O. Pujol, et al., Traffic Sign Detection and Recognition System, in: Springer Briefs in Computer Science, Springer, London, 2011, pp. 81–93.
- [21] C.G. Keller, C. Sprunk, C. Bahlmann, et al., Real-time recognition of US speed signs, in: IEEE Intelligent Vehicles Symposium, 2008.
- [22] Z. Wang, Z. Tang, X. Zhang, Reflection symmetry detection using locally affine invariant edge correspondence, *IEEE Trans. Image Process.* 24 (4) (2015) 1297–1301.
- [23] Z. Zhu, J. Lu, R.R. Martin, et al., An optimization approach for localization refinement of candidate traffic signs, *IEEE Trans. Intell. Transp. Syst.* 18 (11) (2017) 3006–3016.
- [24] D. Deguchi, M. Shirasuna, K. Doman, et al., Intelligent traffic sign detector: Adaptive learning based on online gathering of training samples, in: IEEE Intelligent Vehicles Symposium, 2011.
- [25] C. Bahlmann, Y. Zhu, V. Ramesh, et al., A system for traffic sign detection, tracking, and recognition using color, shape, and motion information, in: IEEE Intelligent Vehicles Symposium, 2005.
- [26] H.H. Chiang, Y.L. Chen, W.Q. Wang, et al., Road speed sign recognition using edge-voting principle and learning vector quantization network, in: 2010 International Computer Symposium, 2010.
- [27] Y. Gu, T. Yendo, M.P. Tehrani, et al., Traffic sign detection in dual-focal active camera system, *IEEE Intell. Veh. Symp.* 32 (14) (2011) 1054–1059.
- [28] G. Wang, G. Ren, Z. Wu, et al., A robust, coarse-to-fine traffic sign detection method, in: International Joint Conference on Neural Networks, 2013.
- [29] T. Kaser, G. Kramar, D. Nozica, Traffic signs shape recognition based on contour descriptor analysis, in: International Conference on Smart Systems and Technologies, 2016.
- [30] J. Mattioli, Minkowski operations and vector spaces, *Set-Valued Anal.* 3 (1) (1995) 33–50.
- [31] J. Shao, *Mathematical Statistics: Exercises and Solutions*, Springer, New York, 2005.
- [32] M.A. Garcia-Garrido, M. Ocana, D.F. Llorca, et al., Robust traffic signs detection by means of vision and V2I communications, in: International IEEE Conference on Intelligent Transportation Systems, 2011.



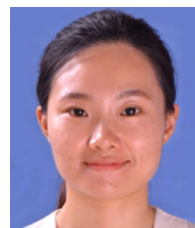
Xianghua Xu is now a professor in the School of Computer Science at Hangzhou Dianzi University, China. He received his Bachelor degree from Hangzhou Institute of Electronic Engineering, and his Ph.D. in Computer Science from Zhejiang University, China. His current research interests include pattern recognition, parallel and distributed computing, and wireless sensor networks. He has published more than 100 peer reviewed journal and conference papers. He is a recipient of the Best Paper Award of IISWC'2012. He is the member of the IEEE, ACM.



Jiancheng Jin is now a graduate student in the School of Computer Science at Hangzhou Dianzi University, China. He received his Bachelor degree from Hangzhou Dianzi University. His current research interest is mainly focus on traffic sign detection in smart traffic systems.



Shanjing Zhang is now an associate professor in the School of Computer Science at Hangzhou Dianzi University, China. He received his Ph.D. in Computer Science from East China Normal University, China. His current research interests in pattern recognition. He has published more than 20 peer reviewed journal and conference papers.



Lingjun Zhang is now an assistant professor in the School of Computer Science at Hangzhou Dianzi University, China. He received his Ph.D. in Computer Science from Tianjing University, China. His current research interests in pattern recognition. He has published more than 10 peer reviewed journal and conference papers.



Shiliang Pu is the director of HIKVISION's Research Institute. He received his Ph.D. in Applied Optics from the University of Rouen in France in 2005. His current research interests include image processing and pattern recognition. He is responsible for the company's technology research and development work in video intelligent analysis, image processing, coding and decoding.

Zongmao Cheng is now an associate professor in the School of Science at Hangzhou Dianzi University, China. He received his Ph.D. in Statistics from East China Normal University, China. His current research interests in pattern recognition and data mining. He has published more than 20 peer reviewed journal and conference papers.